

TABLE II
GROWTH DRIVERS OF THE INDIAN PCB MARKET

Segment	Value	Details
Consumer electronics	US\$ 1670 million+	- Increasing demand for flat TVs (LED and LCD) is primarily driving the growth of this segment. - With government directives to digitise entire pay TV network including cable TV, the set-top box (STB) market has gained significant momentum. - Though only double-sided and multi-layered PCBs are required for STBs, which are within India's capability, PCBs are imported in more than 90 per cent cases. - In spite of having TV production units here, local PCB manufacturers fail to deliver the required volume.
Mobile phones	US\$ 723 million+	88.4 million smartphones are expected to be sold in India in 2015-16. - Leading mobile set manufacturers like Samsung, Micromax, Lenovo, Xiaomi, Karbonn and HTC will be having their manufacturing base in India soon. - Though today's mobile phones need complex, higher layer count PCBs and HDIs that cannot be met by local capability, the immediate opportunity Indian manufacturers can grab is the mobile charger market that needs simple single- and double-sided PCBs.
Computer hardware	US\$ 420 million+	- Despite having a US\$ 5 billion strong PC market, the Indian PCB market has limited opportunities in this segment so far, because no component-level manufacturing is happening in India for computers/PCs. - Apart from PCs, printers, memory cards and USB drives also need PCB in this segment, which have huge domestic production potential. - HP, Lenovo, Samsung and Acer either have their production units here in India or source from EMS. If local PCB manufacturers build in the technical capability of providing higher-layer microvia technology and flexible circuits, the huge domestic market can be captured.
Telecom	US\$ 471 million+	- Rising tele-density in India is leading to higher demand for telecom equipment. By 2020, domestic production of telecom equipment would be US\$ 9.2 billion. - In spite of being one of the large consumers of PCBs, Indian telecom industry is mostly dependent on imports. - The Department of Telecom (DoT) is focusing more on the growth of local manufacturing of telecom equipment that can lead to more opportunities for local PCB manufacturers.
Industrial electronics	US\$ 924 million+	- There has been an increased demand for automation and process control. - Further demand for industrial electronics is expected to be driven by its applications in areas such as artificial intelligence and robotics. - Local PCB manufacturers are supplying the demand for this sector currently; however, need to upgrade their volume and technology to become sole provider in future.
Strategic	US\$ 570 million+	- Almost everything that military and defence uses—starting from weapon to other equipment or even clothing—have very sophisticated electronics integrated on it. And PCB is one basic requirement for all these uses. - Since Indian defence manufacturing has been identified as a priority sector in Make in India initiative and focus is on using indigenous products/components, this would spur more demand for local PCBs.
Automotive	US\$ 490 million+	- The booming Indian automotive industry is driving the demand for automotive electronics. - With increasing digitisation of automobile controls, automotive electronics will continue to grow and so will the market for PCB for auto components. - Electronic devices constitute up to 25 to 30 per cent of auto component costs. This will offer a huge opportunity for automotive PCBs. - Indian manufacturers already have the technical capability for few applications. They need to develop competency for advanced auto applications.
LED lighting	US\$ 840 million+	- Indian LED lighting market is forecasted to reach US\$ 20.9 billion by 2020. - Key growth drivers are: (a) declining LED prices, (b) favourable government initiatives to provide LED lights at subsidised cost and LED installation projects, and (c) increasing awareness. - Indian PCB manufacturers have the capability of meeting complete demand of metal-core PCBs for LEDs. - However, more than 90 per cent requirements are being met through import—local manufacturers are losing out to cheap Chinese PCBs.

However, Indian PCB manufacturers are mostly focused on single-sided, double-sided and multi-layered PCBs with a layer count of four to six, in most cases. A majority of Indian manufacturers adopt the high-mix, medium-volume strategy where different types of PCBs are manufactured in low to medium volumes. There are around 200 PCB

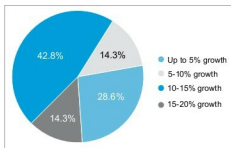


Fig. 1: Expected growth of the Indian PCB industry in FY 2017-18

manufacturers in India—more than 60 per cent of them are very small and unorganised.

All survey participants feel that there is a demand-supply gap in the ecosystem as imports play a significant role in meeting local demand for PCBs. This does not bode well for the economic health of the industry in the long term. Survey